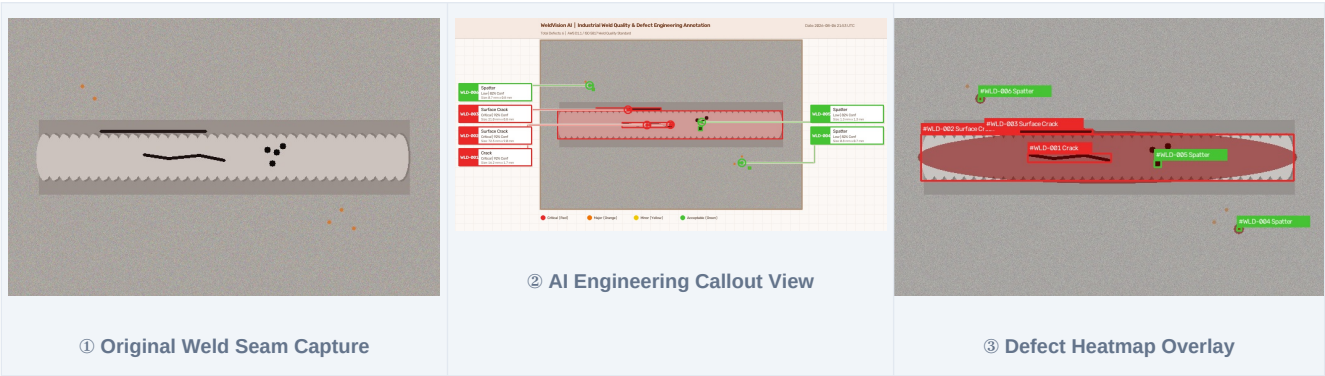


■■ WELD ACCEPTANCE: REJECTED | Quality Score: 55/100 | Defective Area: 4.1% | Defects Found: 6 | Repair Priority: Immediate Repair

WELD QUALITY SCORE	ACCEPTANCE STATUS	OVERALL CONDITION	OVERALL RISK LEVEL	ANALYSIS CONFIDENCE
55 / 100	Rejected	Poor	Critical	91.7%

WELDING INSPECTION VISUAL EVIDENCE



DETECTED WELD DEFECT CATALOGUE

Defect ID	Defect Type	Severity	Weld Zone Location	Size (mm)	Area %	Repair Priority
WLD-001	Crack	Critical	Centerline (Weld Seam)	16.2 mm x 1.7 mm	0.61%	Immediate Repair
WLD-002	Surface Crack	Critical	Centerline (Weld Seam)	72.5 mm x 9.0 mm	2.67%	Immediate Repair
WLD-003	Surface Crack	Critical	Centerline (Left Toe)	21.0 mm x 0.8 mm	0.71%	Immediate Repair
WLD-004	Spatter	Low	Root Pass (Right Toe)	0.8 mm x 0.7 mm	0.03%	No Action Required
WLD-005	Spatter	Low	Centerline (Weld Seam)	1.3 mm x 1.3 mm	0.06%	No Action Required
WLD-006	Spatter	Low	Top Pass (Left Toe)	0.7 mm x 0.8 mm	0.02%	No Action Required

AI INSPECTION VERDICT & QUALITY ASSESSMENT

The inspected weld contains 6 visible defect(s) including Crack, Surface Crack, Spatter. The total estimated defective area is approximately 4.1%. Weld Quality Score is 55/100 with an overall status of 'Rejected'. Immediate repair is required prior to load service.

ENGINEERING CAUSE ANALYSIS & RECOMMENDED REPAIR ACTIONS

POSSIBLE CAUSES

- Excessive thermal stress, high hydrogen content, or rapid cooling rate during solidifying pass.
- High restraint stress, improper joint fit-up, or surface contamination prior to welding.
- High arc voltage, unstable arc, damp electrodes, or improper gas mixture.

RECOMMENDED REPAIR ACTIONS

- Grind out crack completely to sound metal, perform dye penetrant inspection, and re-weld with low-hydrogen electrode.
- Stop-drill ends, excavate defect area completely by gouging/grinding, and execute GTAW repair pass.
- Scrape or chisel spatter droplets off base metal; apply anti-spatter spray pre-weld.

INSPECTION SUMMARY & METRICS

Total Weld Defects	6	Estimated Defective Area	4.1%
Critical Defects	3	Weld Quality Score	55 / 100
Major Defects	0	Weld Acceptance Status	Rejected
Minor Defects	0	Repair Priority	Immediate Repair
Weld Coverage Analyzed	98.3%	AI Inspection Confidence	91.7%

Certified Inspector Name:

Quality Approval Authority:

Inspector Signature:

Date Approved:

[QR CODE
VERIFICATION]

Report ID: WLD-20260806-215320 | Generated by WeldVision AI Quality Engine | Analysis compliant with AWS D1.1 / ISO 5817 weld visual inspection criteria. All defect indications are AI visual estimates and must be verified by a CWI / NDT Inspector.